14. Implement House price Prediction using appropriate machine learning algorithm

Aim

To implement a **House Price Prediction Model** using an appropriate machine learning algorithm in Python and predict the price of a house based on its features.

Algorithm

1. Import the required libraries.
2. Create a dataset containing house features and prices.
3. Separate the input features and target price.
4. Split the dataset into training and testing data.
5. Create a **Linear Regression** model.
6. Train the model using the training data.
7. Predict house prices using the test data.
8. Calculate the R^2 score to evaluate the model.
9. Predict the price of a new house.

Code

```
from sklearn.linear_model import LinearRegression

from sklearn.model_selection import train_test_split

from sklearn.metrics import r2_score


X = [

    [1000, 2, 1],

    [1200, 2, 1],

    [1500, 3, 2],

    [1800, 3, 2],

    [2000, 4, 2],

    [2200, 4, 3],

    [2500, 4, 3],

    [2800, 5, 3],

    [3000, 5, 4],

    [3500, 6, 4]

]
```

```
y = [
    3000000,
    3500000,
    4500000,
    5200000,
    6000000,
    6800000,
    7500000,
    8500000,
    9200000,
    10500000
]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

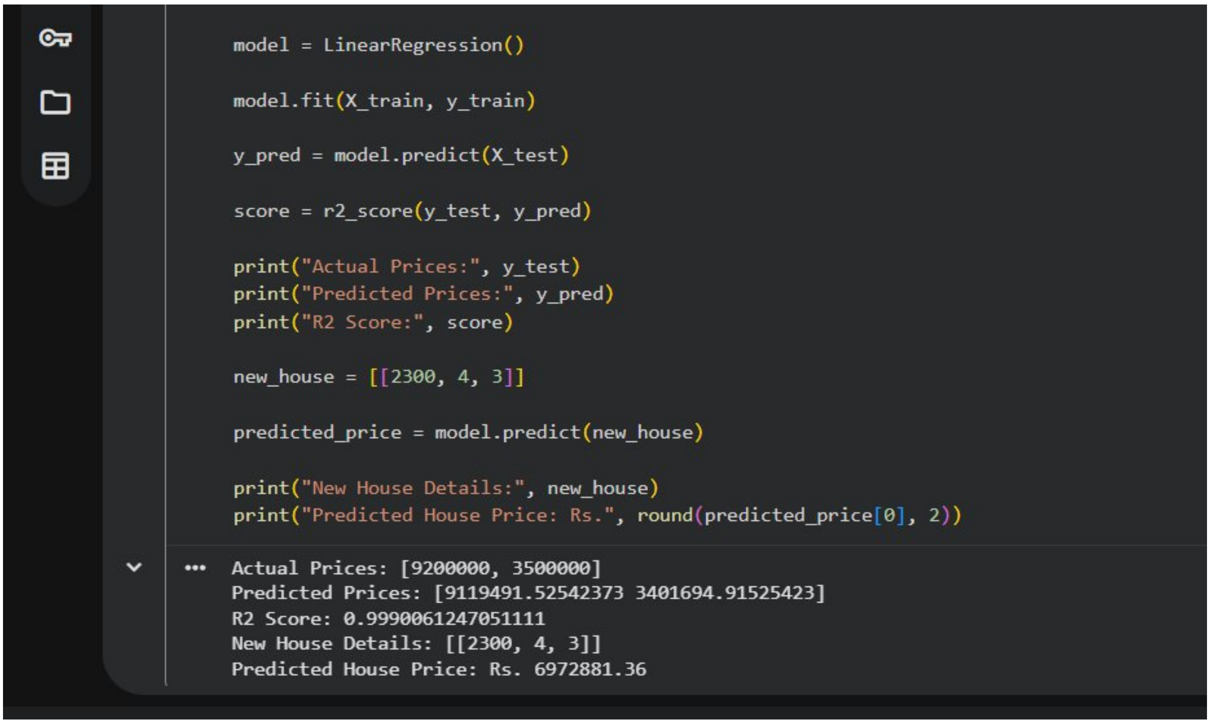
score = r2_score(y_test, y_pred)
print("Actual Prices:", y_test)
print("Predicted Prices:", y_pred)
print("R2 Score:", score)
new_house = [[2300, 4, 3]]
```

predicted_price = model.predict(new_house)

print("New House Details:", new_house)

print("Predicted House Price: Rs.", round(predicted_price[0], 2))

OUTPUT:



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model.fit(X_train, y_train)

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score = r2_score(y_test, y_pred)

print("Actual Prices:", y_test)
print("Predicted Prices:", y_pred)
print("R2 Score:", score)

new_house = [[2300, 4, 3]]

predicted_price = model.predict(new_house)

print("New House Details:", new_house)
print("Predicted House Price: Rs.", round(predicted_price[0], 2))
```

```
Actual Prices: [9200000, 3500000]
Predicted Prices: [9119491.52542373 3401694.91525423]
R2 Score: 0.9990061247051111
New House Details: [[2300, 4, 3]]
Predicted House Price: Rs. 6972881.36
```

Result Statement

Thus, the **House Price Prediction** model was successfully implemented using the **Linear Regression algorithm** in Python, and the price of a new house was successfully predicted.